R. MIRZA
DATA ENGINEER

(910) 518-5312 • Linkedin• mirza.rehanaliya63@gmail.com

PROFESSIONAL SUMMARY

Data Engineer with expertise in designing and developing scalable and efficient data solutions, optimizing **ETL pipelines**, and implementing sophisticated **data workflows** for seamless **data integration, transformation, and analytics**. Skilled in leveraging **Big Data technologies** such as **Apache Spark (PySpark, Spark SQL, Spark Streaming), Hadoop, and Kafka**, along with **cloud platforms** including **Azure (Data Factory, Databricks, Synapse, Data Lake), GCP (BigQuery, Dataflow)** to drive **real-time and batch data processing**.

- Adept at developing efficient data transformation scripts, streamlining ETL pipelines, and automating workflows using **Python (Pandas, NumPy)** with data visualization using Matplotlib.
- Experienced in Python web frameworks, including **Django**, **Flask**, and **Pyramid,** for building scalable and efficient web applications.
- Proficient in writing complex **SQL** queries (joins, window functions, and aggregations) across relational databases, such as **MySQL, PostgreSQL, SQL Server, and Teradata**, ensuring optimal query performance and data quality.
- Skilled in using **Databricks** for collaborative data engineering, leveraging its integration with **Azure** for real-time data analytics and processing.
- Expertise in implementing **Azure Data Factory** to orchestrate data workflows and transfer data between cloud and on-premises environments.
- Skilled in leveraging **Azure services**, including **Azure Blob Storage**, **Synapse Analytics**, and **Data Lake**, for scalable data storage and processing.
- Strong background in **NoSQL databases**, including **MongoDB, Cassandra, DynamoDB, and HBase**, for managing large-scale unstructured data.
- Experienced in creating and managing data models (**Star** and **Snowflake** schemas) and optimizing data pipelines for analytics.
- Skilled in containerization and orchestration tools like **Docker and Kubernetes** for deploying scalable data applications in cloud environments.
- Proficient in developing interactive dashboards and visualizations using **Tableau** and **Power BI** to communicate actionable insights to stakeholders.
- Expertise in version control tools like **Git**, **GitHub**, and **GitLab** to ensure seamless collaboration and CI/CD workflows within agile development teams.
- Experienced in shell scripting and automation using **Bash, Unix shell scripting,** and **Terraform** for infrastructure as code (IaC).

SKILLS

Scripting	Python, SQL, R, Scala, PySpark, Bash, Unix Shell Scripting
Big Data & Processing	Apache Spark (PySpark, Spark SQL, Spark Streaming), Hadoop, Kafka, Hive, HBase
ETL Tools	Databricks, Azure Data Factory, AWS Glue, Apache NiFi
Data Warehouse/ Database	Azure SQL Data Warehouse, MySQL, Microsoft SQL Server, PostgreSQL, Teradata, Snowflake
NoSQL	MongoDB, Cassandra, DynamoDB, HBase
Cloud Services	Azure (Databricks, Data Lake, Data Factory, Blob Storage, Synapse, Functions), GCP (BigQuery, Dataflow)
Data Visualizations	Power BI, Tableau
Version Control	GIT, GitLab, GitHub, Bamboo, Jenkins
Containerization & Orchestration	Docker, Kubernettes

PROFESSIONAL EXPERIENCE

Michael Kors May 2024 -Present
Azure Data Engineer

The goal of this project was to establish a robust, scalable data architecture within Azure, facilitating seamless data integration, transformation, and analysis. Designed and implemented a scalable data pipeline using **Azure Databricks, PySpark, and SQL** to process and analyze sales data in real-time and batch modes. Store and manage data in **Azure Data Lake and Azure SQL Data** Warehouse, enabling advanced analytics and reporting. Visualize key sales insights through interactive dashboards in **Tableau** for data-driven decision-making.

Scripting
- Engineered a robust, scalable **Azure Databricks** architecture for scalable real-time and batch data processing.
- Developed a robust ETL data pipeline using **PySpark** and **SQL** for data transformation tasks.
- Automated large-scale sales data using **PySpark** for seamless integration in **Azure Databricks**.
- Designed fault-tolerant distributed data processing workflows in **Databricks**.

ETL Tools
- Implemented a scalable ETL pipeline in **Databricks** for transforming and enriching sales data.
- Utilized **Azure Data Factory** for orchestrating data ingestion workflows from multiple data sources.
- Developed a scalable ETL pipeline in **Databricks** for real-time data transformation.
- Implemented **Azure Data Factory** to orchestrate seamless data ingestion from multiple sources.
- Automated data cleansing and enrichment workflows using **PySpark** and **SQL** to improve data quality.
- Built monitoring mechanisms in **Databricks** using **Azure Monitor** to track ETL performance and anomalies.
- Designed an efficient batch processing strategy for historical data analysis, leveraging **Azure Data Lake Storage (ADLS)** and **Databricks**
- Established best practices for ETL Framework scalability using **Azure Data Factory**, **Databricks Notebooks**, and parameterized pipelines.

Data Warehouse/Database
- Integrated processed data into **Azure SQL Data Warehouse** for analytics and reporting.
- Optimized data storage and querying using **Databricks' Lakehouse** capabilities.

- Designed partitioning strategies in **Azure Synapse** to enhance query performance for large datasets.
- Implemented Data Modeling best practices using **Star Schema and Snowflake Schema** in **Azure SQL Data Warehouse**.
- Applied performance tuning techniques using **SQL** indexing, caching in **Databricks**, and optimized **Delta Lake** tables to reduce query latency.

Cloud Services
- Designed and deployed solutions with **Azure Data Lake**, **Data Factory**, and **Blob Storage** for scalable data ingestion and storage.
- Used **Azure Functions** for real-time event-driven processing and automation of data workflows.
- Built and managed distributed processing clusters using **Databricks**.
- Leveraged **Azure Synapse Analytics** for advanced analytics and real-time reporting on high-volume transactional data.
- Integrated **Azure Key Vault** for secure credential management in data pipelines and ETL workflows.
- Configured **Azure Monitor** and Log Analytics for real-time performance tracking of data pipelines.

Data Visualizations
- Developed interactive dashboards in **Tableau** to visualize sales trends, product performance, and regional metrics.
- Published business intelligence reports in **Tableau** for stakeholders to drive data-driven decision-making.

Version Control
- Managed the codebase using **GIT** for version control, ensuring efficient team collaboration and change tracking.
- Used **GIT** workflows for branching, merging, and pull requests to maintain a clean and organized repository**.**

Environment
PySpark, Python, Scala, SQL, Databricks, Azure Data Factory, Azure SQL Data Warehouse, Azure Synapse Analytics, Delta Lake, Azure Data Lake, Azure Blob Storage, Azure Functions, Azure Monitor, Tableau, Git.

Blue Cross Blue Shield — Sep 2023 – May 2024
Data Engineer

The goal of this project was to integrate, transform, and analyze customer behavior data by leveraging **SQL Server Integration Services (SSIS)** within a scalable **Azure-based data pipeline**. Developed and deployed a customer analytics pipeline using **Azure Databricks, PySpark, and Kafka** to process clickstream and transaction data in real-time. Managed data storage and governance using **Azure Data Lake and Unity Catalog**, enabling secure, role-based access control and advanced analytics. Built interactive **Power BI dashboards** to uncover insights on customer purchasing patterns and retention strategies.

Scripting
- Designed and implemented **Databricks-based** architecture for real-time and batch customer data processing.
- Developed optimized ETL pipelines in **PySpark** to cleanse and transform large-scale customer data.
- Integrated **Kafka Streams** for processing customer interactions in real-time.
- Enhanced query performance using **Spark SQL** for structured and semi-structured data analysis.

ETL Tools
- Built an **optimized ETL framework** using **SSIS, Azure Data Factory,** and **Databricks** to ingest and process high-volume transactional data.
- Integrated **SSIS packages** with **Azure SQL Database** and **Snowflake** to automate data extraction, transformation, and loading.
- Ensured **data quality and consistency** using **Databricks Delta Live Tables** for automated profiling and anomaly detection.
- Streamlined **Azure Data Factory workflows** for orchestrating data ingestion from **REST APIs, Azure Blob Storage, and on-prem SQL Servers.**
- Optimized ETL job scheduling using **Azure Data Factory, Databricks Job Clusters,** and **Apache Airflow** for efficient resource utilization.
- Established a monitoring framework using **Azure Monitor, Log Analytics, and Power BI** to track ETL performance and system health.

Data Warehouse/Database
- Stored **processed data** in **Snowflake and Azure SQL Database**, ensuring optimized storage and high-performance analytical querying.
- Designed **database schemas using Star** and **Snowflake models** to enhance customer behavior analysis.
- Implemented **Delta Lake on Databricks** for ACID-compliant transactions and efficient data versioning.
- Leveraged **Unity Catalog in Databricks** for fine-grained access control, metadata management, and security enforcement.

Cloud Services
- Deployed **Azure Data Lake** for scalable and secure storage of raw and processed data.
- Configured **Azure Blob Storage** with lifecycle policies for cost-effective data archiving and retrieval

- Implemented **Azure Functions and Event Grid** for serverless automation of data workflows.
- Used **Azure Key Vault** to manage secrets and credentials securely across ETL pipelines.

Data Visualizations

- Developed **interactive Power BI dashboards** to visualize customer engagement trends and purchasing behavior.
- Designed reports to highlight purchasing patterns and retention strategies.

Version Control

- Managed the codebase using **Git** for seamless collaboration and change tracking across teams.
- Established CI/CD workflows to streamline deployment and updates.

Environment:
PySpark, SQL, Kafka, Databricks, Unity Catalog, SSIS, Azure Data Factory, Snowflake, Delta Lake, Azure Data Lake, Azure Blob Storage, Azure SQL Database, Azure Functions, Power BI, Git, Azure DevOps, Azure Monitor, Apache Airflow.

Axis Bank **Aug 2021 – Dec 2022**
Big Data Engineer

The goal of this project was to build a scalable big data platform for analyzing IoT sensor data collected from manufacturing equipment. Designed and implemented a data pipeline using **Apache Kafka**, **PySpark**, and **Hadoop** to process streaming and batch data. Leveraged **Azure HDInsight** and **Azure Data Lake** for large-scale data storage and analytics, enabling predictive maintenance and performance optimization for manufacturing equipment.

Scripting

- Designed a **PySpark**-based framework to process structured and unstructured IoT sensor data.
- Developed real-time ingestion workflows using **Apache Kafka** and **Spark Streaming**..
- Wrote complex **PySpark** scripts for anomaly detection and trend analysis in sensor data.
- Enhanced query performance using **Spark SQL** for structured and semi-structured data analysis.

ETL Tools

- Built a scalable **Kafka**-based ETL pipeline for real-time data ingestion and processing..
- Orchestrated batch workflows using **Azure Data Factory** and **Hadoop** for large-scale data processing.
- Automated ETL workflows using **Apache Airflow** for scheduling and monitoring.
- Integrated **Databricks Delta** for ACID-compliant storage and optimized queries.
- Improved data quality using **Databricks Delta Live Tables** for automated profiling.
- Designed a fault-tolerant ETL strategy for historical and real-time data integration.

Data Warehouse/Database

- Stored **processed data** in **Snowflake and Azure SQL Database**, ensuring optimized storage and high-performance analytical querying.
- Designed **database schemas using Star** and **Snowflake models** to enhance customer behavior analysis.
- Implemented **Hive** for querying semi-structured data from **HDFS**.
- Optimized **SQL** queries with indexing and partitioning techniques.
- Used **HBase** for fast lookups and updates of time-series sensor data.

Cloud Services

- Utilized **Azure HDInsight** to set up scalable **Hadoop** and **Spark** clusters, enabling efficient processing of IoT sensor data across multiple nodes for both batch and real-time analytics, while ensuring high fault tolerance.
- Leveraged **Azure Data Lake Storage (Gen2)** to serve as a centralized repository for raw and processed IoT data, ensuring high scalability, data security, seamless integration with analytics tools, and optimized storage costs.
- Configured **Azure Event Hub** to handle high-throughput ingestion of IoT sensor data streams in real-time, ensuring low latency, reliable message delivery, and seamless downstream integration with processing tools like **Spark Streaming**..
- Developed serverless **Azure Functions** to automate real-time processing of IoT events, such as anomaly detection or generating alerts, triggered by specific thresholds or conditions, reducing the need for manual intervention.
- Used **Azure Key Vault** to securely store and manage sensitive information such as API keys and connection strings, ensuring compliance and secure integration with other Azure services for pipeline workflows.
- Utilized **Azure Monitor** and **Log Analytics** to track system performance, diagnose pipeline bottlenecks, and ensure optimal resource utilization for consistent operational performance, with automated alerts for critical events.

Data Visualizations

- Developed **interactive Power BI dashboards** to display equipment performance trends.
- Published reports showing predictive maintenance insights for stakeholders.

Version Control

- Managed source code using **Git** and established CI/CD workflows in **Jenkins**.

- Automated version control using **GitLab** for team collaboration.

Environment:
PySpark, Hadoop, Kafka, Azure HDInsight, Azure Data Factory, Snowflake, Hive, HBase, Azure Event Hub, Azure Data Lake, Azure Functions, Power BI, Git, Jenkins, Apache Airflow.

Flipkart Jun 2020 – Mar 2021
Python Developer

The goal of this project was to build a real-time data analytics platform for tracking social media sentiment using **Python, GCP** services, and natural language processing tools. Developed and deployed a cloud-native pipeline to collect, process, and analyze social media posts in real-time. Leveraged **BigQuery, Dataflow, Pub/Sub, and Google Looker** for scalable data processing and advanced analytics while utilizing Python libraries such as **SpaCy** and **NLTK** for sentiment analysis.

Scripting
- Designed and implemented pipelines using **Python** and **Apache Beam** for stream processing.
- Developed text processing workflows using **SpaCy** and **NLTK** for sentiment analysis.
- Automated data workflows in **GCP** using **Python** scripts.
- Enhanced batch processing workflows using **PySpark** for scalability.

ETL Tools
- Used **GCP Dataflow** for real-time and batch transformation of social media data.
- Ingested data from APIs into **BigQuery** using **Pub/Sub** and **Dataflow**.
- Designed ETL workflows for REST API integration with **Cloud Functions**.
- Enhanced data quality checks using **Dataflow SQL** for anomaly detection.
- Orchestrated workflows using **Apache Airflow** for efficient job scheduling.
- Stored raw data in **Cloud Storage** with automated lifecycle policies for cost optimization.

Data Warehouse/Database
- Stored processed data in **BigQuery** with optimized partitioning and clustering.
- Designed Star schemas to support fast analytics on sentiment trends.
- Enhanced query performance with **SQL** window functions and caching.
- Built data lineage and governance workflows using **Data Catalog**.
- Implemented **Delta Lake** for ACID compliance in historical data

Cloud Services
- Configured **Pub/Sub** for real-time data ingestion from social media platforms.
- Used **Cloud Functions** for event-driven automation of workflows.
- Deployed **Dataflow** pipelines for distributed processing of large datasets.
- Monitored system health using **Cloud Monitoring** and **Logging** dashboards.

Data Visualizations
- Built interactive dashboards in **Google Looker** to visualize sentiment trends.
- Published reports showing key social media metrics for campaigns.

Version Control
- Managed code using **GitHub**, ensuring proper branching and collaboration.
- Automated CI/CD workflows in **GitHub Actions** for seamless deployment.

Environment:
Python, Apache Beam, GCP (BigQuery, Dataflow, Pub/Sub, Cloud Functions, Cloud Storage, Secret Manager), SpaCy, NLTK, Google Looker, GitHub, GitHub Actions.